Hi Paul,

Here's where the fork landed. The learnings since we split off, written down while they're still accurate. The short version is one line you already half-believe: for a fixed small model, a harness guarantee beats a prose rule, every time. Everything below is a version of that.

The shape we settled on. We stripped the worker back to almost nothing. Vanilla Pi on the floor, a thin worker that's only a command tool, and every behaviour on top added as one explicit remedy you can read and measure. No bundle. The presets aren't an input, they're the result: whichever units, added one at a time from the floor, make a tier pass. Every claim has a unit behind it you can switch off and re-measure.

Easy is won by guidance, not guards. Three prompt lines. Name the layout, work as a careful developer, verify every requirement and run the tests. Took easy from 0/3 to 6/6. The guards we'd reached for fired zero times. On that tier they were weight, not safety, and the lesson stuck: read `guard_blocks` before you believe a guard helped.

The worker's "done" can't be trusted. You'll remember it self-exits on its own weak test. The Phase-4 id it assigns in the route passes the happy path and fails direct construction. A better sentence didn't fix it; a stronger prompt hit the full pass on only half the reps, a coin-flip. A frozen acceptance test did: one that re-runs at settle and hands back the failing one, the gate it can't talk its way past. 3/3 on the ladder at ~43 requests. The guarantee closed what the prose couldn't.

One pass beats the architect loop we built on the fork, with one fixed model. One agent building the whole section in a single pass came in about five times cheaper on requests than the plan-delegation loop, at the same score or better. ~11 requests against 52–64 on easy, ~70 against 300-plus on medium. The split isn't wrong; it's overhead here. It pays when a smart planner hands to a cheaper builder (JetBrains' Junie Eco, 48–57% cheaper), which is a price gap, not the decomposition. That loop earns its keep on a spec too big for one context. The rung we haven't reached, and the one that could flip this.

The real killer on medium was a live-server spiral. Confused by a 303 that read as 200, the worker started a server and curled localhost to see for itself, and the server it started disagreed with the in-process test client, so it curled again, and thrashed to the deadline. The old foreground guard waved the background form through. A guard that refuses any server start and any loopback curl took it from 1-of-3 reaching the gate to 3-of-3.

Thinking spends the turn. Mellum has no effort dial, so the only lever is the prompt. Telling it to reason in one short sentence per action made each step denser, so a turn cap that starved medium (median 10) fit it (median 16). A setting that won on easy lost on medium first. Tier by tier, nothing transfers for free.

The flow carries a phased build, but its gate is only a proxy. Per-phase frozen gates and a git checkpoint per pass kept a seven-phase hard build from breaking an early phase late. The catch: all seven gates passed while the hidden grader failed 5 of 34. The examiner's test is a proxy; the hidden grade is the oracle, and a run isn't done because its own examiner says so.

Two that aren't the model's fault. A LAN serving host handed reasoning back as speech on 236 of 244 turns. We repair it at the agent, but the right move is to pin the serving stack. And the local M3 Max on Q4 matches a rented A6000 on Q4, so we measure for free and only rent for Q8 and up.

Bottom line is, the direction's settled enough to write down and still accurate about what it hasn't measured. Every number's a screen on llama.cpp, a lead, not a rate. The one run that would change the story is a spec past four phases, where one pass might finally run out of room. That's where I'd point the ladder next.

Andre